



A Level • Cambridge (CIE) • Chemistry

39 mins

21 questions

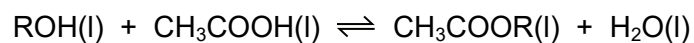
Multiple Choice Practice

Topical Practice MCQ

7. Equilibria

Carefully selected past paper questions by topic.

- 13 An alcohol, ROH, reacts reversibly with ethanoic acid to produce an ester.

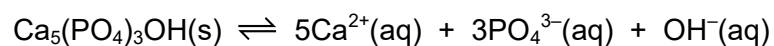


3.0 mol of ROH, 2.0 mol of ethanoic acid and 1.0 mol of water are mixed together. At equilibrium, 1.5 mol of CH₃COOR is present.

What is the value of the equilibrium constant, K_c , for this reaction?

- A** 0.20 **B** 0.25 **C** 2.00 **D** 5.00

- 13 When some solid $\text{Ca}_5(\text{PO}_4)_3\text{OH}$ is added to a beaker of water, an equilibrium is set up.



Which compound, when added to the equilibrium mixture, increases the amount of $\text{Ca}_5(\text{PO}_4)_3\text{OH}(\text{s})$ present?

- A** NH_3 **B** NH_4Cl **C** $\text{CH}_3\text{CO}_2\text{H}$ **D** NaCl

- 14 Hydrogen iodide is added to an evacuated reaction vessel. The vessel is sealed and warmed. A decomposition reaction occurs. Hydrogen and iodine are formed. Some hydrogen iodide remains.

When equilibrium is established, the total pressure is 1.20×10^5 Pa. The partial pressure of hydrogen is 4.00×10^3 Pa.

Hydrogen iodide, hydrogen and iodine are all gaseous under the conditions used.

What is the value of K_p for the equilibrium at this temperature, assuming the decomposition is the forward reaction?

- A 1.19×10^{-3} B 1.28×10^{-3} C 1.38×10^{-3} D 1.43×10^{-3}

- 13 When an equimolar mixture of H_2 and I_2 react, the mole fraction of HI in the final mixture is x .

What is the equilibrium constant, K_p , for the reaction?

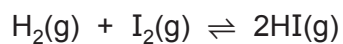
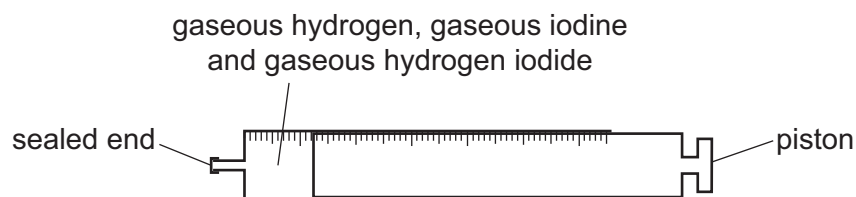
A $\frac{x^2}{(1-x)^2}$

B $\frac{x^2}{(1-2x)^2}$

C $\frac{4x^2}{(1-x)^2}$

D $\frac{4x^2}{(1-2x)^2}$

- 17 The diagram shows a gas syringe with a free-moving piston. The syringe contains gaseous hydrogen, gaseous iodine and gaseous hydrogen iodide at equilibrium.



Three changes are listed.

- 1 increasing the total pressure by adding an inert gas and keeping the volume constant
- 2 increasing the pressure by adding more gaseous hydrogen iodide and keeping the volume constant
- 3 decreasing the volume by pushing the piston to the left

Which changes will result in an equilibrium position at which the rate of the forward reaction has increased?

- A** 2 only **B** 1 and 2 **C** 1 and 3 **D** 2 and 3

26 NaOH(aq) is added to NH_4Cl (aq). The mixture is warmed.

The gas that is produced turns damp red litmus paper blue.

Which row is correct?

	behaviour of the ammonium ion in NH_4Cl	behaviour of the water present on the litmus paper
A	Brønsted–Lowry acid	Brønsted–Lowry base
B	Brønsted–Lowry acid	Brønsted–Lowry acid
C	Brønsted–Lowry base	Brønsted–Lowry acid
D	Brønsted–Lowry base	Brønsted–Lowry base

- 13 When an equimolar mixture of H_2 and I_2 react, the mole fraction of HI in the final mixture is x .

What is the equilibrium constant, K_p , for the reaction?

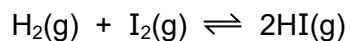
A $\frac{x^2}{(1-x)^2}$

B $\frac{x^2}{(1-2x)^2}$

C $\frac{4x^2}{(1-x)^2}$

D $\frac{4x^2}{(1-2x)^2}$

- 12 Hydrogen and iodine react to form hydrogen iodide in an exothermic reaction. The equation is shown.

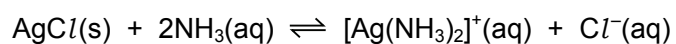


A 1 m^3 reaction vessel contains H_2 , I_2 and HI gases at equilibrium. The temperature is changed such that the total pressure in the 1 m^3 vessel doubles.

What is the effect on the value of K_p and on the position of equilibrium?

	effect on the value of K_p	effect on the position of equilibrium
A	decreases	moves left
B	increases	moves right
C	no change	moves left
D	no change	no change

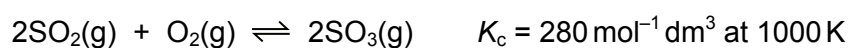
- 6 The equation for the reaction between silver chloride and aqueous ammonia is shown.



What are the units of K_c for this reaction?

- A** no units **B** $\text{mol}^{-1} \text{dm}^3$ **C** mol dm^{-3} **D** $\text{mol}^2 \text{dm}^{-6}$

11 The reaction between sulfur dioxide and oxygen is reversible.



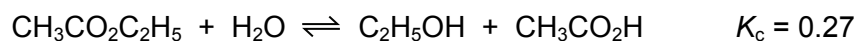
In an equilibrium mixture at 1000 K the sulfur trioxide concentration is 6.00 mol dm^{-3} .

The sulfur dioxide concentration is twice the oxygen concentration.

What is the sulfur dioxide concentration?

- A $0.175 \text{ mol dm}^{-3}$
- B $0.254 \text{ mol dm}^{-3}$
- C $0.318 \text{ mol dm}^{-3}$
- D $0.636 \text{ mol dm}^{-3}$

10 Ethyl ethanoate undergoes the following reaction.



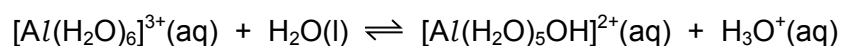
Equal amounts of ethanoic acid and ethanol were mixed together and allowed to reach equilibrium.

At equilibrium, the concentrations of both ethanoic acid and ethanol were 0.42 mol dm^{-3} .

What is the concentration of ethyl ethanoate at equilibrium?

- A 0.22 mol dm^{-3}
- B 0.65 mol dm^{-3}
- C 0.81 mol dm^{-3}
- D 1.54 mol dm^{-3}

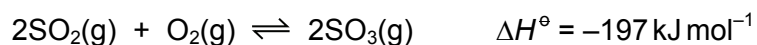
11 Hydrated aluminium ions undergo the following reaction.



Which statement about this reaction is correct?

- A $H_2O(l)$ and $[Al(H_2O)_5OH]^{2+}(aq)$ are a conjugate acid-base pair.
- B $H_2O(l)$ is acting as an acid as it is donating H^+ ions.
- C If $OH^-(aq)$ is added, the equilibrium will move to the right.
- D K_c varies as the pH is varied.

10 A reaction involved in the Contact process is shown.

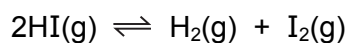


The reaction is investigated at 200 kPa and 700 K and the value of the equilibrium constant, K_p , is found to be Y. The reaction is then investigated at 1000 kPa and 700 K and the value of K_p is found to be Z.

Which statement comparing Y and Z is correct?

- A Y and Z are the same.
- B Y is greater than Z.
- C Z is 2.2 times greater than Y.
- D Z is 5.0 times greater than Y.

- 9 Hydrogen iodide dissociates into hydrogen and iodine.

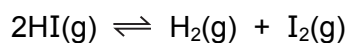


In an experiment, b mol of hydrogen iodide were put into a sealed vessel at pressure p . At equilibrium, x mol of the hydrogen iodide had dissociated.

Which expression for K_p is correct?

- A $\frac{x^2}{(b-x)^2}$ B $\frac{x^2 p^2}{(b-x)^2}$ C $\frac{x^2 p^2}{4b(b-x)}$ D $\frac{x^2}{4(b-x)^2}$

- 9 Hydrogen iodide dissociates into hydrogen and iodine.

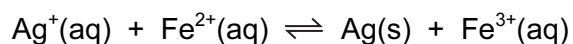


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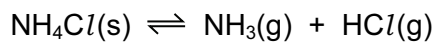
- 9 An aqueous solution was prepared containing a mixture of 1.0 mol of AgNO_3 and 1.0 mol of FeSO_4 in 1.00 dm^3 of water. When equilibrium was established, there was 0.44 mol of $\text{Ag}^+(\text{aq})$ in the mixture.



What is the numerical value of K_c ?

- A** 0.62 **B** 1.40 **C** 1.62 **D** 2.89

- 10 When solid ammonium chloride dissociates at a certain temperature in a 0.500 dm^3 container, ammonia and hydrogen chloride are formed.



The initial amount of ammonium chloride was 1.00 mol , and when the system had reached equilibrium there was 0.300 mol of ammonium chloride.

What is the numerical value of K_c for this reaction under these conditions?

- A** 0.490 **B** 1.63 **C** 1.96 **D** 3.27

- 6 One mole of phosphorus(V) chloride, PCl_5 , is heated to 600 K in a sealed flask of volume 1 dm^3 . Equilibrium is established and measurements are taken.

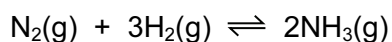


The experiment is repeated with one mole of phosphorus(V) chloride heated to 600 K in a sealed flask of volume 2 dm^3 .

How will the measurements vary?

- A The equilibrium concentrations of $\text{PCl}_3(\text{g})$ and $\text{Cl}_2(\text{g})$ are higher in the second experiment.
- B The equilibrium concentration of $\text{PCl}_5(\text{g})$ is lower in the second experiment.
- C The equilibrium concentrations of all three gases are the same in both experiments.
- D The value of the equilibrium constant is higher in the second experiment.

- 9 Nitrogen reacts with hydrogen to produce ammonia.



A mixture of 2.00 mol of nitrogen, 6.00 mol of hydrogen, and 2.40 mol of ammonia is allowed to reach equilibrium in a sealed vessel of volume 1 dm^3 under certain conditions. It was found that 2.32 mol of nitrogen were present in the equilibrium mixture.

What is the value of K_c under these conditions?

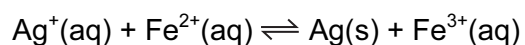
A $\frac{(1.76)^2}{(2.32)(6.96)^3}$

B $\frac{(1.76)^2}{(2.32)(6.32)^3}$

C $\frac{(2.08)^2}{(2.32)(6.32)^3}$

D $\frac{(2.40)^2}{(2.32)(6.00)^3}$

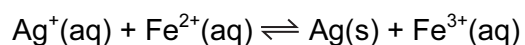
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Answer Key

No.	Paper	Topic	Answer
1	9701_s24_qp_11	7. Equilibria	D
2	9701_w24_qp_12	7. Equilibria	A
3	9701_s23_qp_12	7. Equilibria	B
4	9701_w23_qp_11	7. Equilibria	C
5	9701_w23_qp_12	7. Equilibria	D
6	9701_w23_qp_12	7. Equilibria	B
7	9701_w23_qp_13	7. Equilibria	C
8	9701_m22_qp_12	7. Equilibria	A
9	9701_m20_qp_12	7. Equilibria	A
10	9701_s19_qp_11	7. Equilibria	D
11	9701_s19_qp_12	7. Equilibria	C
12	9701_w19_qp_12	7. Equilibria	C
13	9701_s17_qp_12	7. Equilibria	A
14	9701_w17_qp_11	7. Equilibria	D
15	9701_w17_qp_13	7. Equilibria	D
16	9701_s16_qp_11	7. Equilibria	D
17	9701_s16_qp_12	7. Equilibria	C
18	9701_s15_qp_11	7. Equilibria	B
19	9701_w13_qp_11	7. Equilibria	A
20	9701_w11_qp_11	7. Equilibria	D
21	9701_w11_qp_13	7. Equilibria	D